Trading on News the Moment It Is Released: An Intraday Test of Sentiment Signals in US Stocks

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1. Hypothesis Pitch

When a company gets clearly good or bad news, the market does not finish pricing it in at once. In the minutes and hours after the headline is released, a stock's abnormal return should keep moving in the direction of the news, by more than it costs to trade: up after clearly positive news, down after clearly negative news.

This informs the strategy: the moment a headline about a stock is released, score it with FinBERT, a language model trained on financial text. If the score is strongly positive, buy the stock at the next available minute; if strongly negative, short it. Hold for 60 minutes, then exit, and aim to profit from the part of the reaction that has not happened yet.

Most research on news sentiment uses daily data, where an article can only be dated to the day. That cannot show whether anything is left to trade in the minutes after a headline appears, because the price may have moved before the trade could be placed. This study tests the trade directly. It uses the exact release time of each article from Alpaca's news feed (Benzinga articles) and Alpaca's 1-minute price bars, so every trade is entered after the release and uses only prices available at that moment.

The mechanism is gradual information diffusion and limited attention. The tone of a story is soft, qualitative information that is harder to price than a number such as an earnings surprise. Tetlock, Saar-Tsechansky and Macskassy (2008) find that negative words in firm-specific news predict returns, with prices under-reacting at first, and Chan (2003) finds that prices keep drifting after bad headlines. Attention is also limited and uneven (Barber and Odean, 2008), so not every investor reads every headline at the same moment. The evidence is not one-sided: Tetlock (2007) finds that media pessimism can reverse rather than drift. In liquid US large caps, where algorithms read the same feeds within seconds, the reaction may well be finished before a retail-speed trade can be placed. That is the question this study tests.

The parameters were locked in a written pre-registration before any news or price data was downloaded. There is an in-sample period, and a final year is held out and used once. The window runs from 1 January 2024 to 30 September 2026. The in-sample period is January 2024 to September 2025 (21 months) and the holdout is October 2025 to September 2026 (12 months). This window was chosen because no result from it had ever been seen.

What would prove this wrong, fixed in advance:

- The in-sample coefficient of the 60-minute abnormal return on the FinBERT score is not positive and significant at 5%. That would mean the news is already priced in by the time a trade can be placed, or that any leftover reaction cannot be told apart from noise.
- The 60-minute strategy, after a 10 basis point round-trip cost, does not beat the equal-weight benchmark in the held-out year on the Sharpe ratio. A reaction that exists but is smaller than the cost is not a strategy.

The hypothesis is supported only if both conditions hold. Anything else is reported as not supported.

2. Data Pipeline

This section describes the data the study uses. Section 3 explains how the sample was built from it.

News. The news comes from Alpaca's news API, which carries Benzinga articles. Each article has a headline, the list of tickers it tags and the exact time it was released, to the second in UTC. The window holds 675,087 articles over 33 months.

Prices. Prices are Alpaca's 1-minute bars from the IEX exchange feed, which is the free tier and covers only part of total market volume. A minute with no trade on IEX has no bar at all. Daily bars from May 2023 are used to estimate betas, and SPY is the market.

Sentiment models. FinBERT gives each headline the probability of positive minus the probability of negative, a score between -1 and +1. VADER, a simple word-list scorer, is the comparison model. Only the headline is scored.

Final sample. The sample is 25,049 events (article-ticker pairs) across 50 stocks: 16,852 in sample and 8,197 in the holdout. NVDA (2,035), AAPL (1,641) and AMZN (1,562) have the most articles and no stock has fewer than 300. The mean FinBERT score is +0.14. A total of 27.6% of events score at or above +0.5 and 13.0% at or below -0.5, giving 6,910 long and 3,267 short signals at the 0.5 threshold. FinBERT and VADER agree only weakly (correlation 0.25).

Data limits. The IEX feed carries only a share of all trades, so prices are approximate and thin stocks have gaps. Prices are split-adjusted but not dividend-adjusted, so both benchmarks leave out dividends equally. Two days (23 December 2024 and 10 March 2025) were treated as non-trading days because SPY had fewer than 50 IEX bars on them, so they are missing from every series. Half-day sessions have fewer than 390 bars.

3. Methodology

This section walks through the study in the order it was actually run, from the first decision to the final result, with the number of articles or tickers left after each step.

3.1 From the Window to the Pre-registration

The first step was the pre-registration. The window and split (Section 1) and every rule of the strategy were written down before any news or price data existed. From then on, any change had to be recorded as a dated deviation, with a note of whether it was made before or after seeing returns (Section 5.1).

3.2 Collecting the News

Alpaca's news API returns articles in pages of 50, oldest first. A script paged through the whole window and saved the articles to disk as it went, keeping a note of where it had got to, so that a closed laptop or a failed request meant resuming rather than starting again. It retried on rate-limit and server errors and stopped at once on a refused key. After removing duplicate article ids and anything outside the window, 675,087 articles remained.

3.3 Sifting the Articles

The news was then reduced in four steps, using only the text and times of the articles, never a price. Each article that tags several tickers is split into one row per article-ticker pair, and each pair is one possible event.

- **Roundups removed.** 70,898 articles (10.5%) tag more than two tickers. These are market summaries and lists of movers that say nothing about any one stock, so they are dropped.
- **Tickers cleaned.** Tickers containing a slash, longer than five letters or not purely alphabetic (share classes, warrants, foreign listings) are dropped, as are a fixed list of exchange-traded funds. After this and the roundup rule, 602,334 article-ticker pairs across 10,237 tickers remain.
- **Regular hours only.** Only articles released Monday to Friday from 09:30:00 to 15:58:59 New York time are kept, so that an entry bar exists before the close. An after-hours article could only be traded at the next open, hours or days later, which would bring back the delay this study is designed to remove. This leaves 243,496 pairs (40.4%) across 8,257 tickers.
- **Eligibility.** A ticker needs a kept article in at least 25 of the 33 months and at least 300 kept articles in total. 935 tickers meet the first test, 85 meet the second, and 84 meet both.

3.4 Choosing the Stocks

Only 84 tickers are eligible, so the universe is really just the most heavily covered large US companies. Eighty of them were drawn in random order with a fixed seed (20261027), using only article counts, and the first 50 with usable price data were to form the universe. Before any price was downloaded, two tickers in the first 50 were removed because they did not fit the intent of the rule: FXI, an exchange-traded fund missing from the exclusion list, and GOOGL, a second share class of the company already drawn as GOOG. VGK, another fund, was removed from the reserve list. The next tickers in the draw order took their places, without a redraw.

3.5 Collecting the Prices

SPY was downloaded first, because the days on which it has regular-hours bars define the trading calendar (687 trading days from 2 January 2024 to 30 September 2026), and a day with fewer than 50 SPY bars is treated as a holiday. Then each ticker in the draw order was downloaded, one at a time, as 1-minute bars (split-adjusted) and daily bars. After each ticker, four coverage rules fixed before any bars were downloaded were checked, using bar counts and dates only, never prices or returns: a bar on at least 95% of trading days, at least 25% of the 390 minutes filled on the median day, at least one bar in each of the first and last five trading days, and at least 120 daily bars before 2024 so that betas exist. The run took about 1,200 requests and stopped as soon as 50 tickers had been accepted. Every accepted ticker had bars on 100% of trading days, and the median share of minutes filled ranged from 35% (COST) to 100% (NVDA). Only BLK failed, with bars in a median of 17.9% of minutes, and RBLX, the next ticker on the reserve list, took its place.

3.6 Scoring the Headlines

Only the headline of each article is scored, because the headline is what a trader sees at the moment of release. The final 50 stocks have 25,049 events but only 20,047 distinct headlines, because articles tagging two tickers appear twice, so each headline is scored once and the score is shared. FinBERT took 27 seconds on a laptop GPU and VADER 3 seconds. Scores are stored on disk and reused when the pipeline is re-run.

3.7 Matching Events to Prices

Each event was matched to its entry and exit bars and given an abnormal return using the rules in Section 4.1. An entry bar was found for 24,125 of the 25,049 events. The other 924 are 244 events on days with no trading session and 680 where no bar was available within 5 minutes of the release. The median beta was 1.21.

3.8 Testing the Code Before Looking at Any Return

Before any real return was looked at, the whole notebook was tested on made-up data. Thirteen groups of hand-made cases check the timing rules to the second (for example, a release at 10:00:30 enters at the 10:02 bar, a bar exactly 5 minutes late is allowed and one 6

minutes late is skipped), the 16:00 close and early-close days, the one-position rule, the cost arithmetic and the beta calculation. Twelve simulated stocks, with between 2% and 30% of minutes missing, were used to check that a planted effect of 60 basis points per unit of score is recovered (the estimates were 55 at 60 minutes and 60 at 120 minutes), that four worlds with no effect give nothing (60-minute coefficients between -0.2 and +1.9, all with p-values above 0.35), and that a world in which the stock moves before the release gives nothing to a trader who enters after it. The real-data cells of the notebook could not run until all of these passed.

3.9 Running Once and Reporting Everything

The notebook was then run once on the real data with no change to any rule or code. Every horizon, both periods, all three costs, both models and all thresholds are reported, whatever they show, and only the 60-minute result decides the verdict. The funnel from articles to signals is summarised below:

Step	Count left
Articles downloaded in the window	675,087
Article-ticker pairs after removing roundups, odd tickers and funds	602,334 (10,237 tickers)
Released in regular hours	243,496 (8,257 tickers)
Tickers meeting both eligibility rules	84
Tickers drawn in random order	80
Final universe (49 drawn tickers plus RBLX)	50 tickers, 25,049 events
Events with an entry bar available	24,125
Signals (score of +0.5 or above, or -0.5 or below)	10,177 (6,910 long, 3,267 short)

Table 1: From articles to signals.

4. Strategy and Backtest

4.1 Strategy and Backtest Mechanics

Each article-ticker event is scored. A score at or above +0.5 is a long signal and at or below -0.5 is a short signal. The entry is the open of the first 1-minute bar that starts at least 60 seconds after the release time, so a trade can never use a price that existed before the article was out. The exit is the open of the bar that starts 60 minutes after the entry bar, or the 16:00 close if that comes first. The hold is therefore shorter than 60 minutes for articles released late in the afternoon.

A stock can have only one open position, so a new signal on a stock already held is ignored. A trade is skipped, and counted, if its entry bar starts more than 5 minutes after the release or its exit bar starts more than 5 minutes after the target exit time (Deviation 3). A missing minute is not filled with an old price for entry or exit. A skipped trade never opens a position, so it does not block a later signal on the same stock.

A stock's return over the holding period is mostly the market moving, so for the event study the return is converted to an abnormal return: the stock's return minus its beta times SPY's return over the same minutes. Beta is estimated from the 119 daily returns that end the day before the event day, so it never uses the day it is applied to. The strategy's own profit and loss is the raw stock return in the direction of the signal, not hedged against the market.

A trading cost of 10 basis points round trip is deducted from every trade. Twenty and 30 basis points are tested as sensitivity checks. Capital is split equally across the trades entered on a given day, with no leverage, and days with no trade earn zero. Sharpe ratios assume a zero risk-free rate and are annualised with the square root of 252. The benchmarks are an equal-weight buy-and-hold of the 50 stocks and SPY.

Because new trades were entered on almost every trading day, this layout charges a full 10 basis points to capital each day, however briefly a trade is open. Over 436 trading days in sample that adds up to about 44% of capital in simple terms. It was fixed in advance, but it means the net numbers are dominated by costs, and the before-cost numbers are the better read of whether there is any signal.

4.2 Results

Six metrics are reported:

- **Cumulative return**: the total percentage gained or lost over the period.
- **Annualized return**: the cumulative return expressed as a compound annual rate, so that periods of different lengths can be compared.
- **Sharpe ratio**: return earned per unit of volatility taken on.
- **Maximum drawdown**: the worst peak to trough loss along the way.

- **Sortino ratio**: like the Sharpe ratio, but it only penalises downside volatility.
- **Win rate**: the share of days that were profitable.

Both benchmarks are shown alongside the 60-minute strategy: an equal-weight holding of all 50 stocks and SPY. All results below are for the 60-minute strategy, the only one that decides the verdict.

4.2.1 In Sample Comparison

In sample (January 2024 to September 2025, 436 trading days, 5,060 trades), the comparison is:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with costs)	-39.8%	-25.4%	-5.04	-39.7%	-6.61	33.9%
Strategy (no costs)	-6.8%	-4.0%	-0.68	-11.5%	-0.89	51.1%
Equal-weight stocks	85.8%	43.1%	1.69	-22.2%	2.38	56.9%
SPY	44.3%	23.6%	1.39	-16.8%	1.80	57.3%

Table 2: In-sample performance comparison.

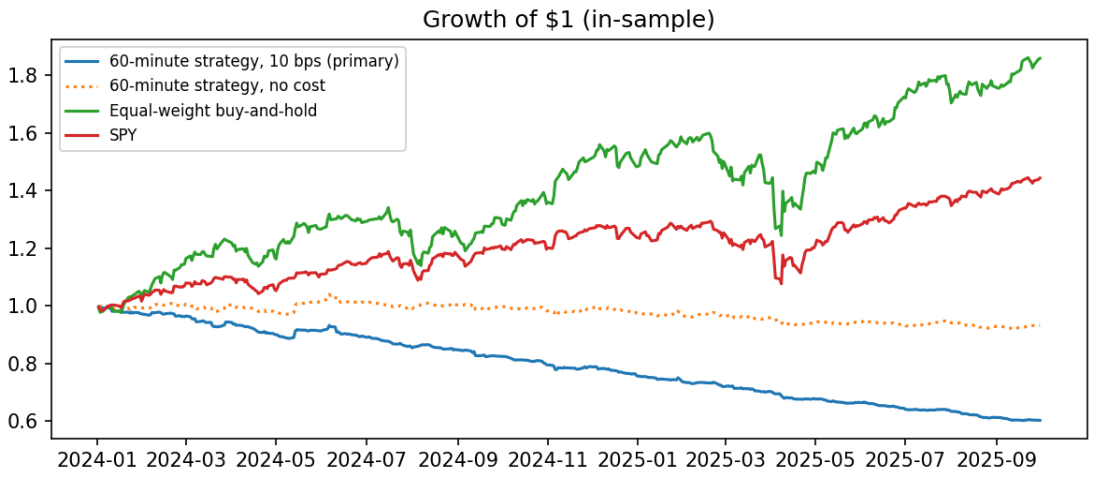


Figure 1: Growth of $1, in-sample.

In sample the strategy lost 39.8% after costs and 6.8% before costs, with Sharpe ratios of -5.04 and -0.68. The equal-weight stocks gained 85.8% (Sharpe 1.69) and SPY gained 44.3% (Sharpe 1.39) over the same period. Almost all of the net loss is the daily cost: before costs, the strategy was close to flat and slightly negative, which is what the event study in Section 5.2 predicts.

4.2.2 Out of Sample Comparison

In the holdout (October 2025 to September 2026, 251 trading days, 3,070 trades), the comparison is:

Series	Cumulative Return	Annualised Return	Sharpe Ratio	Max Drawdown	Sortino Ratio	Win Rate
Strategy (with costs)	-17.8%	-17.8%	-3.74	-17.8%	-5.68	33.5%
Strategy (no costs)	5.7%	5.7%	1.09	-3.2%	1.68	50.2%
Equal-weight stocks	23.2%	23.3%	1.18	-12.2%	1.99	51.0%
SPY	14.5%	14.5%	1.11	-9.1%	1.64	52.6%

Table 3: Holdout performance comparison.

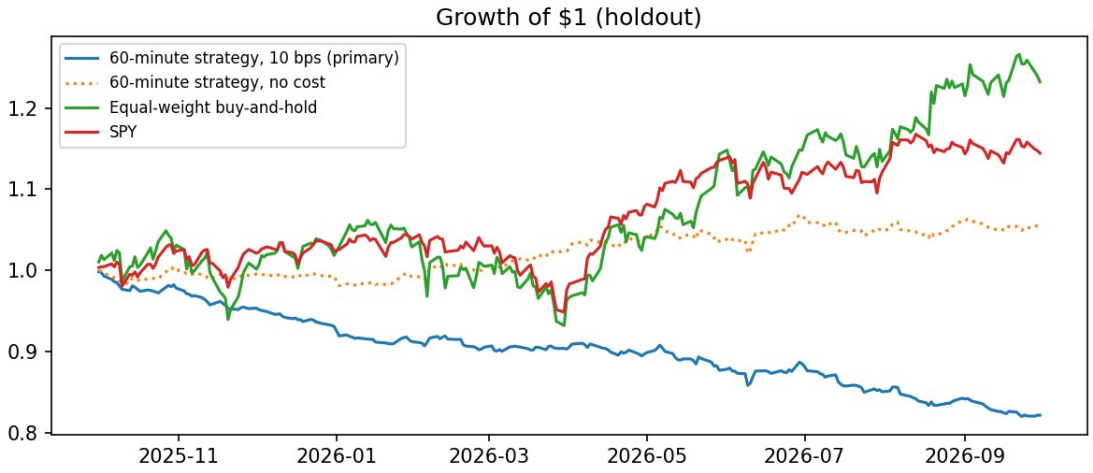


Figure 2: Growth of $1, holdout.

The strategy lost 17.8% after costs and gained 5.7% before costs. Its net Sharpe ratio was -3.74, against 1.18 for the equal-weight stocks and 1.11 for SPY. Before costs the Sharpe ratio was 1.09, which is still below the 1.18 of simply holding the stocks. The decision rule would therefore fail on the second condition even if trading were free.

In one sentence: the 60-minute strategy lost money after costs in both periods, and even before costs it did not beat the equal-weight benchmark in the holdout.

Decision rule. Condition 1 (the in-sample 60-minute coefficient positive and significant at 5%) is not met: the coefficient is +3.95 basis points per unit of score and the p-value is 0.56. Condition 2 (holdout net Sharpe above the equal-weight benchmark) is not met: -3.74 against 1.18. The hypothesis is not supported.

5. Methodology Defence

5.1 Pre-registered Parameters

The parameters locked on 27 October 2026 are in Table 4, followed by the four dated deviations in Table 5. All four were made before any return was seen.

Parameter	Value, as locked
Window	2024-01-01 to 2026-09-30; in sample to 2025-09, holdout 2025-10 to 2026-09
Universe	50 tickers: seeded draw of eligible tickers, first 50 with usable bars
Articles kept	At most 2 tickers tagged; regular hours only (09:30:00 to 15:58:59 New York)
Signal	FinBERT headline score of +0.5 or above long; -0.5 or below short
Entry	Open of first 1-minute bar starting 60 seconds or more after release
Primary hold	60 minutes, or the 16:00 close if earlier
Comparison holds	5, 15, 30, 120, 180, 240 and 300 minutes (exploratory)
Cost	10 bps round trip; 20 and 30 bps as sensitivity checks
Abnormal return	Stock return minus beta times SPY return; beta from 120 daily closes
Portfolio	Equal weight across the day's trades, no leverage, one position per stock
Primary test	OLS of 60-minute abnormal return on score, standard errors clustered by ticker
Decision rule	In-sample coefficient significant at 5% AND holdout net Sharpe above equal-weight

Table 4: Locked parameters.

Deviation	What changed	When
1	Regular-hours articles only (after-hours ignored)	Before any data
2	FXI and GOOGL removed from the draw order, VGK from the reserve list; no redraw	After article counts, before bars
3	Bar-coverage rules and the 5-minute staleness rule fixed	Before bars were downloaded
4	Six clarifications (event-study sample, unhedged P&L, beta, entry to the second, one-position rule, holidays and early closes)	After validation, before real returns

Table 5: Dated deviations.

5.2 Testing the Mechanism

The primary test regresses the 60-minute abnormal return, in basis points, on the FinBERT score of every article-ticker event that has a valid trade, with standard errors clustered by ticker. Each unit of score is the full range from clearly negative to clearly positive. The coefficient is the average extra abnormal return, in basis points, per unit of score:

Period	Events	Coefficient (bps per unit)	Std. error (ticker)	p-value (ticker)	p-value (date, robustness)
In sample (decides)	15,815	3.95	6.73	0.558	0.427
Holdout (descriptive)	8,094	4.01	2.45	0.101	0.136

Table 6: Primary test, 60-minute horizon, FinBERT.

The in-sample coefficient is +3.95 basis points per unit of score (p = 0.56), which is positive but a tenth of the 10 basis point cost and not statistically different from zero. The holdout gives +4.01 (p = 0.10) and is not used for the decision. Clustering by date instead of by ticker, which allows for several articles on the same day moving together, gives the same answer (p = 0.43 in sample). The same regression on holds that were not cut short by the close gives +4.46 (p = 0.55). Condition 1 is not met.

The same regression was run at every horizon. These results are exploratory, because testing eight horizons in two periods means some will look significant by chance:

Horizon (min)	In sample: coef.	Std. error	p-value	Holdout: coef.	Std. error	p-value
5	1.27	1.93	0.509	1.94	0.66	0.003
15	3.14	3.65	0.390	2.42	1.33	0.070
30	3.90	5.11	0.445	3.05	1.99	0.126
60 (primary)	3.95	6.73	0.558	4.01	2.45	0.101
120	6.69	5.81	0.249	7.03	3.48	0.043
180	8.33	6.70	0.214	5.97	4.03	0.138
240	6.70	5.68	0.238	5.87	4.42	0.184
300	6.63	5.15	0.198	5.64	4.64	0.223

Table 7: Event study by horizon (FinBERT, basis points per unit of score). Only the 60-minute row decides the verdict.

All 16 coefficients are positive, which is the sign the hypothesis predicts, but they are small: 1 to 4 basis points for the first hour and 6 to 8 basis points from two hours onwards. Only two of the 16 have a p-value below 0.05, the holdout at 5 minutes (+1.9 basis points, p = 0.003) and at 120 minutes (+7.0, p = 0.043). With 16 tests, one or two such results are what chance alone would produce. The 5-minute estimate is tight, but at under 2 basis points it is far below the cost.

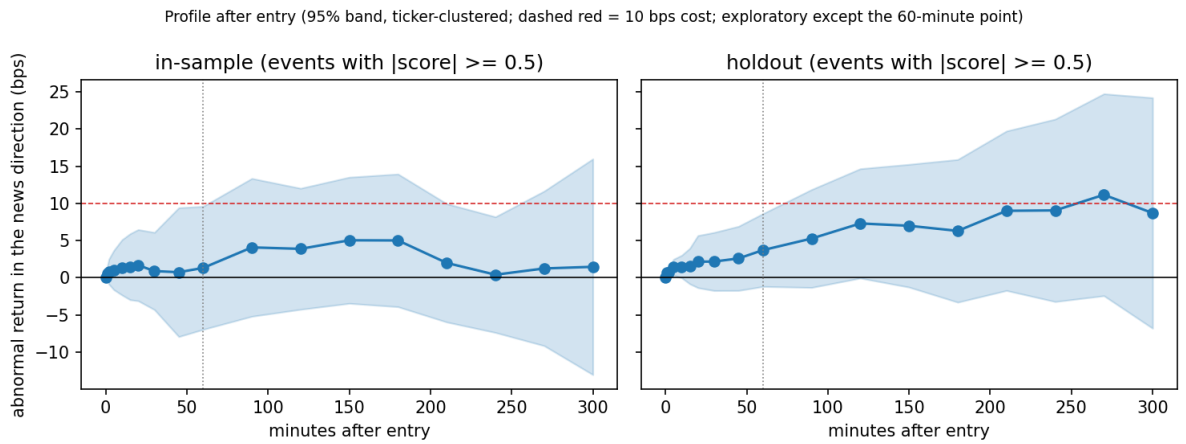


Figure 3: Average abnormal return in the direction of the news at each minute after entry, for signals of 0.5 or more (95% band, clustered by ticker; dashed line is the 10 bps cost).

The profile above shows the whole path. In the first minute after entry the average signal had moved about 0.6 basis points in the news direction, and by 20 minutes about 2. In sample it never gets near the 10 basis point cost line, and at 240 minutes it is back near zero. In the holdout it drifts higher, to roughly 9 to 11 basis points at 210 to 270 minutes, but the band is wide (the lower edge is below zero throughout) and the drift is not there in sample. It is a lead to test on new data, not a result.

5.3 Threshold Sensitivity

The 0.5 threshold was fixed in advance, so the other thresholds are exploratory checks of whether the verdict depends on that choice:

Model and threshold	In sample: trades	Net Sharpe	Holdout: trades	Net Sharpe	Holdout gross Sharpe
FinBERT 0.3	6,137	-5.41	3,626	-4.50	1.16
FinBERT 0.5 (primary)	5,060	-5.04	3,070	-3.74	1.09
FinBERT 0.7	3,798	-4.42	2,399	-2.09	2.16
VADER 0.3	3,595	-3.30	1,900	-3.86	0.06
VADER 0.5	1,474	-1.33	781	-0.34	1.28
VADER 0.7	549	-1.34	262	-0.83	0.21

Table 8: Sensitivity to the signal threshold and the sentiment model (60-minute hold, 10 bps cost).

Net Sharpe ratios are negative for every threshold and model in both periods, so the verdict does not depend on 0.5. FinBERT at 0.7 has a holdout gross Sharpe of 2.16, the best of any variant, but it is one holdout year, it was never an in-sample result (-0.83 in sample before costs), and it is a lead only.

5.4 Sentiment Model Choice

FinBERT was chosen in advance because it is trained on financial text, and VADER is the simple comparison. They disagree a lot: VADER is exactly zero on 49.1% of headlines and reaches +0.5 or above on only 7.6%, so at the same thresholds it makes about a quarter of FinBERT's trades (2,542 against 10,177). Its 60-minute coefficients are -7.3 in sample ($p = 0.12$) and -0.2 in the holdout ($p = 0.97$), so it carries no signal either. At 0.5 its net Sharpe ratios (-1.33 in sample and -0.34 in the holdout) look better than FinBERT's only because it trades on fewer days and so pays less daily cost, not because its signals are better. Its threshold was not tuned to match FinBERT's trade count.

6. Pitfall Checklist

6.1 Look-ahead Bias

Every trade uses only what was known at its entry: the entry bar starts after the release, the exit uses only later bars, and the beta uses only earlier days (rules in Section 4.1). A worked example is saved with the results: a Piper Sandler note on Snowflake (score +0.86) was released at 10:22:52 New York time on 2 January 2024, so the earliest entry was 10:23:52, the entry bar started at 10:24:00 at a price of 191.44, the exit bar started at 11:25:00, and the beta used closes up to 29 December 2023. The notebook also contains a test that changes every bar after a trade's exit and confirms that the trade does not change.

Reading the headline is the one assumption that cannot be checked. Benzinga's timestamp is when the article appeared there, so if a faster source had it earlier, the true reaction began before our entry. That would make our results look worse than a trader with faster access would see, not better.

6.2 Survivorship Bias

The universe is drawn from tickers that exist and have price data now, and only 84 tickers qualified, so it is made up of well-known, heavily covered companies. Stocks that were delisted during 2024 to 2026 and have no minute data are not in the sample, which may flatter the results slightly. In this case the effect is probably small, since the result is a null. One ticker, DJT, trades in the data from the start of 2024 under its current name, but before March 2024 its news was about the earlier shell company it merged with. It was kept because the coverage rules were fixed in advance and changing a ticker after seeing which one passed would be a form of hindsight.

6.3 Overfitting/ Data Snooping

The 60-minute hold, the 0.5 threshold, the cost and the decision rule were fixed before any data was seen, the notebook was tested on simulated data (Section 3.8) before it was allowed to see real returns, and the real run was made once with no code changes. The seven other horizons, the extra thresholds, VADER and the 20 and 30 basis point costs are all labelled exploratory. Eight horizons in two periods is 16 tests, and two of them having p-values below 0.05 is about what chance would give.

6.4 Transaction Costs/Slippage

The 10 basis point round trip is a rough figure. For liquid large caps it is probably fair for small orders, but it ignores the spread at the moment of a news release, which is usually wider than average, and it ignores the cost of borrowing stock to short. Because prices are IEX only, the real execution price could also differ from the bar's open. At 20 and 30 basis points the 60-minute holdout net Sharpe ratio falls from -3.74 to -8.57 and -13.40. No horizon

has a before-cost holdout Sharpe ratio that is clearly above the equal-weight benchmark in both periods, so lower costs would not rescue the result.

Horizon (min)	Holdout Sharpe, 0 bps	10 bps	20 bps	30 bps
5	1.40	-17.29	-35.98	-54.67
15	0.37	-9.72	-19.82	-29.91
30	0.90	-6.14	-13.18	-20.22
60	1.09	-3.74	-8.57	-13.40
120	0.49	-3.01	-6.51	-10.01
180	0.42	-2.73	-5.88	-9.03
240	0.81	-2.08	-4.98	-7.87
300	1.13	-1.60	-4.32	-7.05

Table 9: Holdout Sharpe ratio by horizon and cost (exploratory except the 60-minute row at 10 bps).

6.5 Out-of-Sample Validation

The holdout year (October 2025 to September 2026) was run once, after every rule was locked. It gave the same answer as in sample: a coefficient of about +4 basis points at 60 minutes ($p = 0.10$, not significant) and a net Sharpe ratio well below the benchmark. Nothing from the holdout was used to design or change anything. Two of the exploratory holdout results (5 and 120 minutes) were significant at 5%, and two of 16 tests is what chance gives. There is also an openly stated limit on the other side of the ledger: the in-sample and holdout periods are consecutive, so a regime change in 2026 would not have been tested by either.

Skipped trades and truncated holds

The 5-minute staleness rule matters because the free feed has gaps. At 60 minutes there were 10,177 signals. Of these, 8,130 were executed, 1,749 were ignored because the stock already had an open position, and 298 (3.5% of live signals) were skipped: 53 on non-trading days, 180 with a stale entry bar and 65 with a stale exit bar. Of the executed trades, 766 (9.4%) were cut short by the 16:00 close. The truncated share rises with the horizon, from under 1% at 5 minutes to 67% at 300 minutes, which means the longer holds are increasingly an afternoon strategy with a shorter real hold, and should be read with that in mind:

Horizon (min)	Signals	Executed	Ignored (open position)	Skipped (stale or no session)	Cut short by the close
5	10,177	9,426	433	318	0.6%
15	10,177	9,022	826	329	2.0%
30	10,177	8,634	1,224	319	4.4%
60	10,177	8,130	1,749	298	9.4%
120	10,177	7,498	2,384	295	22.3%
180	10,177	7,085	2,805	287	35.8%
240	10,177	6,790	3,125	262	49.1%
300	10,177	6,600	3,347	230	66.7%

Table 10: Trades by outcome and horizon (all periods).

7. Efficiency Touchpoint

Efficiency here means how much it costs, in time and in data access, to run the strategy, and how fast it can react. Once the data is on disk the analysis is cheap: loading, aligning and computing abnormal returns for all eight horizons took about 1.5 seconds, the event study 0.6 seconds, and a full set of backtests across all horizons and costs 0.13 seconds. Scoring 20,047 headlines with FinBERT took 27 seconds on a laptop GPU. The slow parts were collecting data: the news download was a long run of resumable calls, and the bars took about 1,200 calls over roughly 15 minutes. Every step can be resumed after an interruption, so nothing has to be repeated.

Speed is the bigger efficiency question. A trade at the open of the bar after a one-minute delay is slow compared with how quickly the market can react to a headline, and the first-minute and first-five-minute coefficients (about +0.6 and +1.3 to +1.9 basis points) suggest that most of the reaction is over almost at once. A faster feed would be needed to compete for it, and the cost of a faster feed and faster execution has to be less than a move of about 2 basis points per unit of score. So the bet is on data and infrastructure, not on the model.

The free data is a limit on efficiency too. IEX carries only a share of market volume, so thin stocks have gaps, and the 5-minute staleness rule drops a small share of signals (Section 6.5). A paid, consolidated feed would remove that.

8. Risk Touchpoint

The strategy is invested almost all of the time, in a small number of positions at once. In sample it entered trades on every one of the 436 trading days, with a median of 11 trades on an active day and at most 27 in one day, and at most 11 open at once. The five most-traded stocks (NVDA, AMZN, AAPL, GOOG and AMD) made up 26.9% of trades, so the strategy is heavily tilted to the most-covered large caps. About 66% of trades were long, and the average trade had a market beta of 0.42, so it carried some market exposure even though it was meant to trade on news.

Single-day losses were modest: the worst day was -2.0% in sample (11 June 2024) and -1.6% in the holdout (9 June 2026), and the best days were +2.5% and +1.3%. The large maximum drawdowns (39.7% in sample, 17.8% in the holdout) are not from a big shock. They come from the 10 basis point cost being charged every day and compounding, which is why the before-cost drawdowns are much smaller (11.5% in sample and 3.2% in the holdout at 60 minutes).

Several risks are not captured by this backtest. Shorting costs and availability were ignored, and short signals were about a third of trades. A 10 basis point cost may be too low in the minutes after a news release, when spreads widen and prices jump. Positions held through a trading halt, or an article released just before a large gap, are not modelled separately. Finally, a larger account would move prices, which small orders in large caps would not.

9. Reflection

The hypothesis is not supported. Neither condition of the decision rule was met: the in-sample 60-minute coefficient was +3.95 basis points per unit of score ($p = 0.56$) when it needed to be significant, and the holdout net Sharpe ratio was -3.74 against 1.18 for the equal-weight benchmark. The sign is the one the hypothesis predicts, at every horizon and in both periods, but the size is a few basis points, well below the cost of trading it. The result covers a narrow set: 50 well-covered US large caps, 2024 to 2026, free IEX price bars, Benzinga headlines only, and a 60-minute hold fixed in advance.

The result is consistent with a market that prices a headline within seconds to minutes, faster than a trade placed one minute after the release can act. The coefficients are positive at every horizon, so the news does carry a little information after it is released: about +1 to +4 basis points in the first hour. But that is well below the cost of trading it. For heavily covered US large caps the answer is no, and every rule behind that answer was fixed before the data was seen.

The leads worth carrying forward are small. In the holdout only, the average move in the news direction kept rising to about 9 to 11 basis points over three to four and a half hours, and FinBERT at 0.7 had a before-cost holdout Sharpe of 2.16. Neither appeared in sample, neither survives the cost, and both were found by looking at exploratory outputs, so they would need a new period and different stocks before anyone believes them.

If this experiment were conducted again, I would begin by measuring the first seconds directly. Some extensions I would add:

- **Faster data and execution**: a consolidated, tick-level feed would test the seconds after release, where most of the move appears to happen, and remove the gaps of the IEX feed.
- **Pre-release movement**: measuring the return in the minutes before each article would show how much of the move had already happened, and whether news is moving prices or prices are prompting news.
- **Realistic costs**: a spread and size model for the minutes after a release, with borrowing costs on shorts, instead of a flat 10 basis points.
- **Smaller and less covered stocks**: the universe here is the most heavily covered large caps, where the market is fastest. Mid and small caps may react more slowly, though they are also costlier to trade.
- **Fresh data for any lead**: the late-afternoon drift in the holdout and the 0.7 threshold would need a later period to check them.
- **Better signals**: scoring the full article text, separating first reports from repeats, and weighting by news type (earnings, analyst notes, product news) might sharpen what is a very noisy headline-only score.

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